# Restaurant Startup Studio

## Checkpoint 4: User Study Report

Evaluation & Full Report

IS492 — Spring 2026

University of Illinois at Urbana-Champaign

Team: Alex Chen, Laura Shih, Mubtasim Raad, Takumi Nishi

Date: April 30, 2026

# 1. Study Overview

This report presents results from a user study conducted on the Restaurant Startup Studio, a multi-agent AI system designed to help first-time entrepreneurs plan a restaurant business in Champaign, IL. The system guides users through a three-step flow (Basics → Ideas → Manual) with specialized agents for regulatory, financial, marketing, and operations guidance.

Two participants with distinct professional backgrounds were recruited to evaluate the tool across four structured tasks. Data was collected via the System Usability Scale (SUS) questionnaire and semi-structured qualitative interviews.

# 2. Participant Profiles

| | Participant A | Participant B |
|---|---|---|
| **Background** | J.D. candidate; currently a Compliance Tax Specialist at KPMG | CFO of a DeFi startup; former investment banking analyst |
| **Expertise** | Regulatory compliance, tax law, entity structuring, audit procedures | Financial modeling, cash-flow forecasting, startup fundraising, risk analysis |
| **Restaurant Exp.** | None — represents a novice entrepreneur | Advised one food-tech client; no direct restaurant ops experience |
| **Relevance** | Evaluates regulatory/legal accuracy and permit guidance depth | Evaluates financial planning rigor and budget breakdown credibility |

# 3. Task Descriptions

Each participant completed the following four tasks using the deployed application at *restaurants-ai-48ke.vercel.app*:

| Task | Instruction Given to Participant | What It Tests |
|---|---|---|
| **Task 1** | "You want to open a Korean dining restaurant in Champaign with a $100K budget targeting college students. Enter your information and generate concept suggestions." | Basics page input, form clarity |
| **Task 2** | "Compare the three generated restaurant concepts and select the one you think is most viable." | Ideas page comprehension, decision support quality |
| **Task 3** | "Generate and review the operations section of your business plan manual." | Manual page, output quality/trust |
| **Task 4** | "Review the results regarding regulatory information." | Regulatory agent, answer accuracy |

# 4. Quantitative Results: System Usability Scale (SUS)

The SUS is a 10-item Likert scale questionnaire (Brooke, 1986) yielding a score from 0–100. Scores above 68 are considered above average usability. Each question is rated on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

## 4.1 Participant A — Compliance Tax Specialist (KPMG)

| Q# | Statement | Rating (1–5) | Response |
|---|---|---|---|
| 1 | I think that I would like to use this system frequently. | 4 | Agree |
| 2 | I found the system unnecessarily complex. | 2 | Disagree |
| 3 | I thought the system was easy to use. | 4 | Agree |
| 4 | I think that I would need the support of a technical person to be able to use this system. | 2 | Disagree |
| 5 | I found the various functions in this system were well integrated. | 4 | Agree |
| 6 | I thought there was too much inconsistency in this system. | 2 | Disagree |
| 7 | I would imagine that most people would learn to use this system very quickly. | 5 | Strongly Agree |
| 8 | I found the system very cumbersome to use. | 2 | Disagree |
| 9 | I felt very confident using the system. | 4 | Agree |
| 10 | I needed to learn a lot of things before I could get going with this system. | 2 | Disagree |

**SUS Score (Participant A):** 77.5 / 100

## 4.2 Participant B — CFO, DeFi Startup

| Q# | Statement | Rating (1–5) | Response |
|---|---|---|---|
| 1 | I think that I would like to use this system frequently. | 3 | Neutral |
| 2 | I found the system unnecessarily complex. | 2 | Disagree |
| 3 | I thought the system was easy to use. | 4 | Agree |
| 4 | I think that I would need the support of a technical person to be able to use this system. | 1 | Strongly Disagree |
| 5 | I found the various functions in this system were well integrated. | 4 | Agree |

| | | | |
|---|---|---|---|
| 6 | I thought there was too much inconsistency in this system. | 3 | Neutral |
| 7 | I would imagine that most people would learn to use this system very quickly. | 4 | Agree |
| 8 | I found the system very cumbersome to use. | 2 | Disagree |
| 9 | I felt very confident using the system. | 3 | Neutral |
| 10 | I needed to learn a lot of things before I could get going with this system. | 2 | Disagree |

**SUS Score (Participant B): 70.0** / 100

## 4.3 SUS Score Summary

| Participant | SUS Score | Interpretation |
|---|---|---|
| A — Compliance Tax Specialist | 77.5 | Above Average |
| B — CFO, DeFi Startup | 70.0 | Above Average |
| **Average** | **73.8** | **Above Average** |

The average SUS score of 73.8 falls **above** the industry benchmark of 68, indicating that the Restaurant Startup Studio demonstrates good perceived usability. Participant A rated the system slightly higher, likely due to the tool's strong regulatory checklist aligning with their compliance background. Participant B's slightly lower score reflects a desire for more granular financial modeling capabilities.

# 5. Qualitative Results: Semi-Structured Interview

After completing all four tasks, each participant answered six open-ended questions. Responses are presented verbatim below.

## Q1: What part of the tool was most useful?

### Participant A (Compliance Tax Specialist)

The regulatory checklist in Section 6 of the manual was the standout feature for me. It referenced specific Illinois and Champaign codes—like Rule 750.4000 for the CUPHD operating permit and Sec. 37-5 for zoning. That level of citation is unusual for an AI tool. I also appreciated that it organized permits chronologically (before buildout, before opening, ongoing), which mirrors how a compliance professional would actually sequence the work. The insurance basics section was a nice addition too—most first-time founders completely overlook product liability coverage.

### Participant B (CFO, DeFi Startup)

The startup budget breakdown in Section 5 was the most valuable part. It pulled in real macro data—CPI-U at +2.4%, bank prime around 6.75%, Champaign County average weekly wages. That kind of anchoring gives a first-time founder actual numbers to stress-test against rather than just guessing. The phase-by-phase playbook in Section 11 was also practical because it tied financial milestones to a real timeline instead of presenting everything as a flat list.

## Q2: What part felt confusing?

### Participant A

The three concept cards on the Ideas page felt somewhat redundant. All three showed the same budget fit range (~$75k–$200k) and each card ended with a long paragraph echoing back my original input verbatim. It wasn't immediately clear what differentiated them beyond the concept name. I had to read carefully to notice the differences in operating hours and target customers. A side-by-side comparison table would have made the decision much faster.

### Participant B

The budget breakdown uses percentage ranges (e.g., "often 25–45%" for build-out) but never gives a calculated dollar figure for my specific $100K budget. That's confusing because the system clearly knows my budget from the Basics page. I expected it to say something like "$25K–$45K for build-out" rather than making me do the math. Also, the "Fine-tuned agents active: 0" badge at the top of the manual is confusing—it made me wonder if the agents were actually working or if I was getting generic output.

## Q3: Did the AI suggestions feel locally grounded?

### Participant A

Yes, more than I expected. The regulatory section cited CUPHD food service sanitation rules, City of Champaign zoning code excerpts, and Illinois Part 750 Food Code. Those are the correct authorities for Champaign—not just generic FDA references. The mention of the April 30 renewal calendar for operating permits is a specific local detail. However, I'd note that the alcohol licensing section was vague and didn't reference Champaign's specific liquor commission process, which is actually one of the more complex local requirements.

### Participant B

Partially. The financial data felt locally grounded—Champaign County Q3 2025 average weekly wage of ~$1,309, compared against Cook County at ~$1,618. That's a useful benchmark. But the competitor overview section used mock placeholders ("Neighborhood staple (mock)," "Trend-forward indie spot (mock)") instead of naming actual Champaign restaurants. For a tool targeting Champaign specifically, I expected it to reference real competitors on Green Street or in downtown Champaign. The marketing section did mention Campustown vs. Downtown corridors, which was a nice local touch.

## Q4: Did the manual feel specific enough for a beginner founder?

### Participant A

For a beginner, it's a strong starting framework. The step-by-step playbook breaks things into four phases with realistic time estimates, which is exactly what someone without business experience needs. The "beginner mistakes to avoid" section at the end is practical and direct. My concern is that the legal disclaimers ("not legal advice, verify with professionals") appear so frequently that a nervous first-timer might lose confidence in the output. The disclaimers are appropriate, but the repetition could be reduced.

### Participant B

It's a solid 70% of what a beginner needs. The structure is excellent—concept summary through pre-opening checklist in a logical order. What's missing is the "so what" layer. For example, the cash-flow section says to "model a downside month with revenue −20–30% from base" but doesn't provide an actual template or example calculation. A beginner wouldn't know how to build that model. Similarly, it mentions food cost assumptions but never suggests a target percentage (e.g., 28–32% for a fast-casual concept). More worked examples would bridge the gap between guidance and action.

## Q5: What information was missing?

### Participant A

Three things stood out: (1) No mention of the Illinois Department of Revenue requirements for sales tax registration, which is mandatory before opening. (2) The alcohol licensing section needs Champaign-specific details—the local liquor commissioner, class of license, and typical processing timeline. (3) No guidance on employer obligations beyond a one-liner—things like Illinois-specific employment posters, OSHA reporting, and the Illinois Wage Payment and Collection Act. These are compliance fundamentals that a first-time employer would miss.

### Participant B

The biggest gap is a financial pro forma. There's no projected P&L, no break-even analysis with actual numbers, and no cash-flow projection template. The tool references macro data (CPI, interest rates, county wages) but never synthesizes it into a usable financial model. I'd also want to see unit economics—average ticket size, covers per hour needed to break even, labor cost as a percentage of revenue. Finally, there's no mention of funding sources (SBA microloans, local grants, angel investors)—just a suggestion to "stress-test at prime near 6.75%."

## Q6: Would you trust this tool for early restaurant planning? Why or why not?

### Participant A

I'd trust it as a structured starting point—definitely more useful than asking ChatGPT a general question about opening a restaurant. The regulatory citations are real and verifiable, which builds credibility. But I'd emphasize to any user that this is a planning scaffold, not a compliance clearance. The permit checklist gives you the right categories to pursue, but you still need a lawyer for entity structuring and a CPA for tax registration. For the price point of free, it's impressive. I'd rate my trust level at about 7 out of 10.

### Participant B

For ideation and early concept validation, yes—I'd trust it. The three-concept comparison is a smart way to force a founder to think through trade-offs rather than fixating on their first idea. For financial planning, I'd trust the structure but not the numbers. The budget percentages are reasonable industry ranges, but they're not personalized to the user's specific situation. No bank or investor would accept this as a business plan without a proper pro forma. I'd say 6 out of 10 on trust—useful for direction, not for decision-making on its own.

# 6. Analysis & Key Themes

## 6.1 Strengths Identified

- **Structured workflow:** The Basics → Ideas → Manual flow was intuitive to both participants. Neither required assistance navigating between steps.
- **Local grounding:** Both participants noted that the tool references Champaign-specific data sources (CUPHD rules, county wage data, Campustown/Downtown corridors), which distinguishes it from generic AI outputs.
- **Regulatory citation quality:** Participant A (compliance professional) confirmed that the cited codes and rules (Rule 750.4000, Sec. 37-5, FDA Food Code sections) are real and correctly attributed.
- **Beginner-friendly language:** The manual avoids jargon and provides actionable phasing with time estimates, which both participants rated positively for the target user.

## 6.2 Weaknesses & Improvement Areas

- **Concept card differentiation:** The Ideas page concepts were perceived as too similar. A comparative table or visual highlighting key differences would improve decision support.
- **Financial depth:** Budget percentages are not converted to dollar amounts based on user input. No pro forma, break-even template, or unit economics are provided.
- **Mock competitor data:** Competitor profiles are labeled as “mock” placeholders rather than referencing real local businesses, which undermines credibility for the market analysis section.
- **“Fine-tuned agents active: 0” display:** This technical badge confused both participants and should either be hidden from the end-user view or replaced with a more user-friendly status indicator.
- **Regulatory gaps:** Missing Illinois sales tax registration, Champaign-specific liquor licensing details, and detailed employer compliance obligations.

## 6.3 Trust Assessment

Both participants expressed conditional trust in the tool. Participant A rated trust at 7/10 (strong for regulatory scaffolding, weaker for legal advice replacement), while Participant B rated trust at 6/10 (strong for concept exploration, weaker for financial decision-making). The average trust rating of 6.5/10 suggests the tool succeeds as a planning accelerator but needs deeper personalization to serve as a standalone planning resource.

# 7. Recommendations for Next Iteration

- **Convert budget percentages to dollar amounts** using the user’s stated budget from the Basics page (e.g., “$25K–$45K for build-out based on your $100K budget”).
- **Add a side-by-side comparison view** on the Ideas page to highlight key differences between generated concepts (hours, customer type, location style).
- **Include a downloadable financial template** (even a simple spreadsheet) with pre-filled formulas for break-even analysis and monthly cash-flow projections.

- **Remove or restyle the “Fine-tuned agents active: 0” badge** to avoid confusing non-technical users.
- **Expand regulatory coverage** to include Illinois sales tax registration, Champaign liquor commission process, and employer-specific compliance requirements.
- **Consolidate legal disclaimers** into a single prominent notice at the top of the manual rather than repeating throughout each section.

# 8. Conclusion

The Restaurant Startup Studio achieved an average SUS score of 73.8, indicating above-average perceived usability across two domain-expert participants. The system’s core value proposition—transforming minimal user inputs into a structured, locally grounded startup manual—was validated by both participants. Key areas for improvement center on deepening financial personalization, improving concept differentiation on the Ideas page, and closing specific regulatory gaps for the Champaign jurisdiction. These findings will inform the final iteration of the system ahead of the CP4 submission.